

This laboratory is comprised of practice exercises with integration by parts, trigonometric integrals, and integral word problems.

Exercise 1.

Compute the following integrals.

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|----------------------------|---------------------------------|------------------------------------|
| 1. $\int x e^x dx$ | 6. $\int x \cos(3x + \pi) dx$ | 11. $\int \ln(x)^2 dx$ |
| 2. $\int x \sin(x) dx$ | 7. $\int x \ln(x) dx$ | 12. $\int e^x \cos(x) dx$ |
| 3. $\int x^2 \cos(x) dx$ | 8. $\int \frac{\ln(x)}{x^2} dx$ | 13. $\int \cos(x) \ln(\sin(x)) dx$ |
| 4. $\int (x + 3)e^{-x} dx$ | 9. $\int x \ln(x + 1) dx$ | 14. $\int \cos(x)^3 dx$ |
| 5. $\int x \sin(2x) dx$ | 10. $\int \ln(x) dx$ | 15. $\int \tan(x)^3 \sec(x) dx$ |

Exercise 2.

Compute the following integrals.

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| 1. $\int_{\ln(2)}^{\ln(3)} x e^x dx$ | 3. $\int_1^2 \frac{\ln(x)}{x^2} dx$ | 5. $\int_0^{\pi/3} \tan(x) dx$ |
| 2. $\int_0^{\pi/2} x \sin(x) dx$ | 4. $\int_0^{\pi} e^x \sin(x) dx$ | 6. $\int_0^{\pi/4} \frac{1}{\cos(x)} dx$ |

Exercise 3.

A water tank ruptures and water leaks out at a rate of $r(t) = 100e^{-\frac{t}{100}}$ liters per minute. Assuming that the tank ruptures at $t = 0$, how much water leaks out during the first hour?

Exercise 4.

The velocity of a particle is given by $v(t) = 2 \cos(\pi t) + t$ centimeters per second at time t . If the particle starts ten centimeters away from you, how far away is the particle after one and a half seconds?

Exercise 5.

A ball is thrown upward with a speed of two meters per second. Assuming a constant gravity of $a(t) = -g$ meters per second square, find at which time t the ball stops moving up and starts moving down. Hint: the time t satisfies $v(t) = 0$.

Exercise 5.

A car on Broadway is moving at thirty kilometers per hour when an ice cream cart starts crossing the street thirty meters ahead. The car driver slams on the breaks and starts accelerating at a constant negative rate of $a(t) = 50000$ kilometers per hour square.

1. How much time does it take for the car to stop?
2. Is there ice cream on the windshield of the car upon stopping?